A Later Test Set Is Not a New Domain: Pretraining Familiarity Survives a Contamination-Free Hold-Out

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Abstract

Time-series foundation models are evaluated almost exclusively on public archives that predate them, so a strong score cannot be separated from having seen the test set during pretraining. The obvious remedy is a hold-out that postdates the models. We build one: thirteen forecasters — four classical, three trained per dataset, six pretrained — on seven groups drawn from five domains, every observation published after the last model was released, and every dataset rebuildable without an API key. Under this protocol pretrained models win 5 of 7 groups, lose one to a Theta baseline, and on daily exchange rates are indistinguishable from a seasonal naive forecast, along with every other method tested.

We then ask what separates the wins from the losses, and report a negative result: the two intrinsic properties one would reach for — seasonal strength and spectral entropy, measured on the input window — do not account for the pattern, and seasonal strength is if anything negatively associated with the advantage. What does track it is corpus familiarity. Our largest gain (28% lower MASE than the best classical method, on weekly Wikipedia pageviews) falls on Wikipedia pageviews, the domain TimesFM's authors describe as the bulk of its pretraining corpus, at the same granularities and differing only in time window. Within the pretrained family, where every model forecasts identical series so that series difficulty cancels, the TimesFM family outranks the Chronos family by -0.53 on Wikipedia against -0.09 everywhere else (1,500 vs. 754 series, Mann–Whitney $p < 10^{-5}$). We conclude that a temporal hold-out removes memorisation of a window but not familiarity with a domain, that benchmarks therefore need domain hold-outs stated relative to disclosed corpora, and that the practitioner's question is less which model is better than whether their domain is one the model was raised on.

1 Introduction

The claim attached to time-series foundation models is that a single pretrained network forecasts unseen series without being fitted to them, and that it does so better than methods estimated on the target series itself. That claim is usually supported by evaluation on a standard collection of public archives. Those archives are the difficulty. Every widely used benchmark set — the M-competition data, ETT, the Monash repository — predates the models being evaluated and is distributed in the same public form that pretraining corpora are assembled from. A high score is therefore consistent with two very different explanations, generalisation and memorisation, and the published evaluations cannot distinguish them.

The remedy is not a better statistic but a better hold-out. This paper evaluates on observations that did not exist when the models were trained. We assemble seven forecasting groups from five domains, every one of them published continuously by an institutional source, and place the entire test window after the release date of the most recently released model in the comparison. Nothing in the hold-out can have been memorised because none of it had happened yet.

Under that protocol the headline finding of the foundation-model literature survives only in part. Pretrained models are clearly better on some domains and clearly not better on others. The interesting question is what separates them, and our answer is not the one we set out to give: we expected a property of the series and found a property of the corpus. The domain on which pretrained models gain most is the domain one of them was largely pretrained on — the same source and the same granularities, differing only in time window — and the advantage is largest for exactly the model whose corpus it is.

This is not memorisation of the test set; the hold-out makes that impossible. It is the weaker but more durable form of the same problem. A model that has read a decade of Wikipedia pageview series has learned what Wikipedia pageview series do, and January 2026 pageviews still do it. Moving the test window forward controls for the first kind of contamination and leaves the second untouched, which means the field's evaluations cannot be repaired by recency alone.

Contributions.

1. **A contamination-free protocol**, with a hold-out that postdates every evaluated model, over five domains built only from keyless, continuously published sources so that any reader can rebuild the exact panel from the fetchers we release.
2. **Two negative results.** Pretrained models do not beat classical baselines everywhere: we identify one domain where a Theta baseline ranks first and one where nothing beats seasonal naive. And the intrinsic explanation fails — neither seasonal strength nor spectral entropy accounts for where the advantage appears.
3. **Evidence that pretraining familiarity persists past a temporal hold-out**, from a within-family comparison on identical series that no property of those series can explain.
4. **Significance rather than decimal places.** We report multiple-comparison rank intervals and corrected paired tests, and show that most published-style gaps between leading pretrained models fall inside them.
5. **Cost alongside accuracy**, measured on identical hardware in the same runs.

2 Related work

Pretrained forecasters. Chronos [Ansari et al., 2024] tokenises scaled series and trains a language-model architecture over the token sequence; Chronos-Bolt and Chronos-2 are its faster successors. TimesFM [Das et al., 2024] is a decoder-only patched transformer trained on a large corpus dominated by web and search traffic. Moirai [Woo et al., 2024] targets arbitrary frequency and multivariate input via masked patch prediction. These models are trained on differing and only partially disclosed corpora, which is exactly why an evaluation cannot verify by inspection that a given test set was excluded.

Benchmarks and their contamination. The M4 [Makridakis et al., 2020] and M5 [Makridakis et al., 2022] competitions established the evaluation conventions this paper follows: MASE and sMAPE against a seasonal naive scale, and multiple-comparison rank intervals rather than raw mean losses. Their data, along with the Monash archive [Godahewa et al., 2021], is public and static, which makes it ideal for comparability and unusable for testing memorisation. Recent work has begun to raise leakage explicitly; our contribution is to make the hold-out temporal and the collection reproducible rather than to audit corpora post hoc.

Classical baselines. Theta [Assimakopoulos and Nikolopoulos, 2000] won M3 and remains difficult to beat on seasonal series; automatic ARIMA and ETS [Hyndman et al., 2008] are the standard automated alternatives. We use the `statsforecast` implementations so that the baselines are the ones practitioners would actually run.

3 Protocol

3.1 Contamination-free hold-out

The forecast origin is fixed at 1 January 2026 for every group; each model receives the observations preceding it as context and is scored on the horizon that follows. Every evaluated checkpoint — `amazon/chronos-bolt-small` and `-base`, `amazon/chronos-2`, `google/timesfm-2.5-200m-pytorch`, `google/timesfm-3.0-pytorch` and `Salesforce/moirai-2.0-R-small` — was published before that date, so no test observation existed when any of them was trained.

We deliberately do not tabulate pretraining cutoffs. Several of these models disclose their corpora only in summary, and a table of dates assembled from release notes would give the protocol a precision it does not have. The argument does not need it: an observation recorded in 2026 cannot appear in a checkpoint published in 2025, whatever that checkpoint's internal cutoff was. What we can pin exactly is the software, and `requirements-lock.txt` records every library version used to produce these numbers.

3.2 Domains

All five domains are published continuously by an institution, are retrievable without registration or an API key, and are not redistributions of an existing benchmark. Table 1 lists the resulting groups.

Wikipedia pageviews (daily, weekly, monthly). Human attention: strongly periodic, heavy-tailed, punctuated by exogenous spikes.

Weather (hourly, ERA5 reanalysis via Open-Meteo). Smooth, dominated by a clean daily cycle.

Air quality (hourly, CAMS via Open-Meteo). Same sampling rate and daily cycle as weather, but spiky and heavy-tailed. Included specifically to separate "good at hourly data" from "good at smooth data".

Electricity (hourly, Danish grid settlement). The domain most often claimed in the foundation-model literature, but evaluated here without the contamination that the ETT datasets carry.

Exchange rates (daily, ECB reference rates). The adversarial case: the standing result in forecasting is that nothing reliably beats a naive forecast, so this domain tests whether a benchmark can detect the absence of an effect.

Series selection is specified before any result is seen — the full city list, the full currency list, the publisher's own column set — and nothing is added or removed afterwards. One mechanical exclusion applies: a series taking fewer than three distinct values across its entire input window is dropped, because MASE divides by an in-sample seasonal-naive error that is then zero and the metric is undefined. Exactly one series in the study meets this condition (a hydro-generation column for a Danish price area with no hydro, reported as zero throughout), and the rule is stated in these terms rather than as a threshold because a relative one cannot catch it: its scale is negligible in absolute terms and large relative to itself.

3.3 Horizons, seasonality and context

Horizons and seasonal periods follow the M4 conventions for the matching frequency, so that numbers here sit beside prior work rather than beside a convention we invented: 14 steps at daily with $m = 7$, 13 at weekly, 48 at hourly with $m = 24$. Three choices depart from that and each is a judgement we state rather than bury.

Weekly seasonality is $m = 1$, not 52. This is M4's own setting for its weekly group, adopted there because a seasonal ARIMA search at $m = 52$ is intractable; it exhausted memory here too. Monthly Wikipedia uses a horizon of 8 rather than M4's 18, because with the origin at 1 January 2026 only eight complete months of hold-out exist — a direct consequence of insisting the test window be recent, and the clearest cost of that insistence.

The hourly groups are capped at 2,016 context points, twelve weeks, applied identically to every model. The ERA5 series run to 17,000 points, longer than anything in M4; no pretrained model can attend to that (their windows stop between 512 and 2,048) and a seasonal ARIMA search over it does not terminate. Truncating in the loader rather than per model keeps the comparison honest: every method sees exactly the same input. Groups with more than 500 series are subsampled to 500 with a fixed seed, which is recorded in the harness.

3.4 Metrics and significance

We report MASE and sMAPE as defined by the M4 organisers, and weighted quantile loss and 80% interval coverage for the probabilistic models. Because a table of mean losses invites ranking by the third decimal place, our primary comparison is by rank: series are ranked across models, and each model receives an average rank with a Nemenyi interval derived from the Friedman statistic, following the multiple-comparisons-with-the-best presentation used to report M5. We additionally test each model against the per-group winner with a Wilcoxon signed-rank test, Holm-corrected across comparisons. We use signed-rank rather than a paired t-test because MASE across a panel is strongly right-skewed, and we avoid the label "Diebold–Mariano" because that test compares errors over time within one series, which is not the comparison a panel benchmark makes.

The gap between a mean and a rank is not academic here, and we met it by accident. Before the exclusion above was in place, the near-constant Danish hydro series gave the two neural models mean MASE values of order 10^6 on the electricity group — one series in forty-seven moving the group mean by six orders of magnitude, while every model's typical behaviour on that group was unremarkable. The rank-based conclusion for that group was identical with and without the series. A benchmark reported as a mean is one degenerate series away from a headline; reported as ranks it is not.

4 Results

4.1 Accuracy

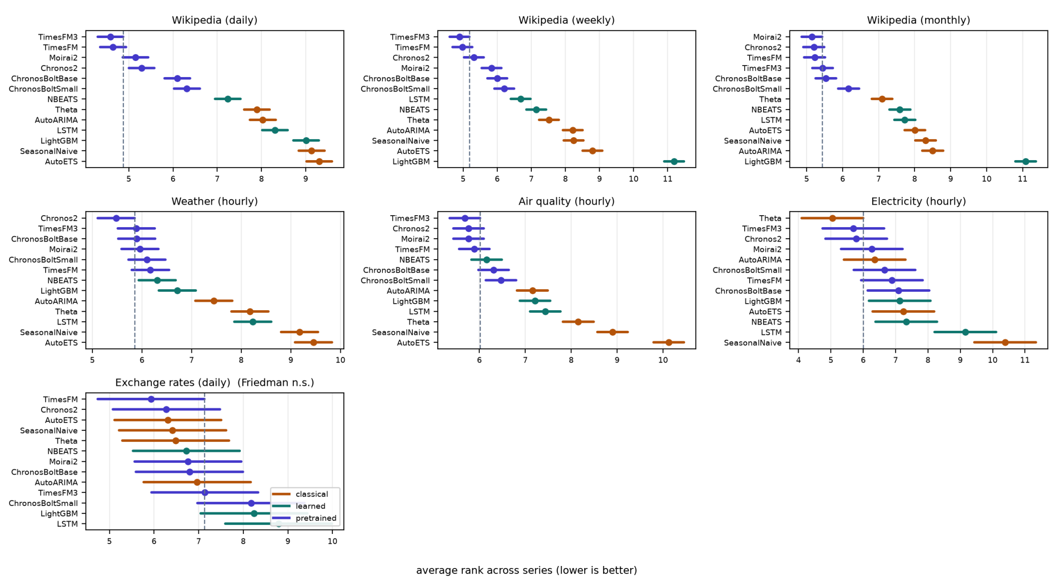

Figure 1: Average rank with Nemenyi intervals, per group. Any model whose interval crosses the dashed line is statistically indistinguishable from the best model in that panel. Exchange rates, on the bottom row, is the panel where every interval overlaps every other and the Friedman test does not reject.

Table 2 is the paper's central result, and it does not read like the tables in the papers introducing these models. A pretrained model holds the top rank in 6 of the 7 groups, but that count overstates the case and we do not use it: in 1 of those the Friedman test does not reject, so the column has a winner in the arithmetic sense only. Counting a rank-first under a null result as a win is precisely the overclaim

Model	Wikipedia (daily)	Wikipedia (weekly)	Wikipedia (monthly)	Weather (hourly)	Air quality (hourly)	Electricity (hourly)	Exchange rates (daily)
Classical							
SeasonalNaive	1.080	1.642	0.822	1.359	1.142	1.443	2.405
Theta	0.961	1.499	0.746	1.280	1.120	**1.045**	2.448
AutoETS	1.149	1.817	0.941	1.558	1.514	1.255	2.468
AutoARIMA	0.928	1.505	0.853	1.191	1.024	1.183	2.502
Trained per dataset							
LightGBM	1.162	3.772	1.545	1.118	0.983	1.260	3.227
LSTM	0.915	1.202	0.786	1.269	0.988	1.376	2.859
NBEATS	0.870	1.326	0.819	1.093	0.903	1.227	2.539
Pretrained							
ChronosBoltSmall	0.790	1.132	0.641	1.067	0.923	1.200	2.614
ChronosBoltBase	0.774	1.143	0.635	1.055	0.915	1.279	2.501
Chronos2	0.734	1.106	**0.620**	**1.043**	**0.895**	1.155	**2.390**
TimesFM	0.695	1.089	0.621	1.073	0.903	1.201	2.400
TimesFM3	**0.689**	**1.073**	0.623	1.069	0.901	1.157	2.418
Moirai2	0.748	1.135	0.629	1.067	0.906	1.185	2.517

Table 1: Mean MASE by group; lower is better, best per column in bold.

Model	Wikipedia (daily)	Wikipedia (weekly)	Wikipedia (monthly)	Weather (hourly)	Air quality (hourly)	Electricity (hourly)	Exchange rates (daily)
Chronos2	5.29	5.31	5.20	5.48	5.76	5.78	6.28
TimesFM3	4.58	4.89	5.44	5.88	5.68	5.70	7.14
TimesFM	4.64	4.98	5.22	6.17	5.89	6.89	5.93
Moirai2	5.15	5.83	5.15	5.96	5.76	6.26	6.76
ChronosBoltBase	6.10	6.00	5.53	5.89	6.31	7.09	6.79
ChronosBoltSmall	6.31	6.21	6.16	6.10	6.47	6.65	8.17
NBEATS	7.24	7.15	7.59	6.31	6.16	7.33	6.72
Theta	7.90	7.51	7.09	8.18	8.15	5.04	6.48
AutoARIMA	8.03	8.21	8.50	7.45	7.15	6.35	6.97
LSTM	8.31	6.69	7.73	8.24	7.44	9.15	8.79
AutoETS	9.31	8.79	8.01	9.46	10.13	7.24	6.31
SeasonalNaive	9.13	8.24	8.30	9.18	8.90	10.39	6.41
LightGBM	9.01	11.19	11.08	6.71	7.21	7.13	8.24

Table 2: Average rank across series (lower is better). Underlined entries are statistically indistinguishable from the best model in that column under the Nemenyi interval.

this paper exists to argue against. The defensible statement is that pretrained models win 5 groups, lose one, and are indistinguishable from everything else — seasonal naive included — on the last.

The two exceptions are the informative ones. On electricity Theta ranks first and the leading pretrained models are merely indistinguishable from it. On exchange rates the tied set contains every method tested; the benchmark detects no difference between any of the thirteen, which is the right answer for that domain rather than a failure of the design.

4.2 The seven groups, one at a time

Wikipedia pageviews (500 series each, daily / weekly / monthly). The strongest result for pretrained models anywhere in the study, and the least surprising once §4.3 is read. TimesFM-3 ranks first on daily and weekly with only two models tied with it; Moirai-2 takes monthly. Every classical method is clearly separated below. These series are bursty — a death, an election or a film release moves a page by an order of magnitude overnight — and a method fitted to the preceding few hundred observations has no way to anticipate the shape of such a burst, whereas a model trained on a decade of them does.

Weather (300 series, hourly). Chronos-2 first, with five models tied. The daily cycle here is close to deterministic, and the interesting observation is that this makes the pretrained advantage *smaller*, not larger: when the structure is clean, a method estimated on the series itself recovers most of it, and there is less left for a prior to supply.

Air quality (379 series, hourly). Same sampling rate and same daily cycle as weather, but spiky, and the pretrained models pull further ahead than they do on weather. This is the comparison the domain was added for, and it rules out the simplest explanation of the weather result — that pretrained models are simply good at hourly data.

Electricity (46 series, hourly). The one group a classical method wins. Theta ranks first, six models are tied with it, and no pretrained model is separated from it in either direction. Load and generation on a national grid are the most regular series in the study: strong daily and weekly cycles, physically bounded, with little of the burst behaviour that Wikipedia has. It is also the domain the foundation-model literature most often claims, and the claim is normally supported on ETT — data from 2016–2018 that sits in every pretraining corpus.

Group	Seasonal strength	Spectral entropy	Pretrained gain
Wikipedia (daily)	0.10	0.93	+25.8%
Wikipedia (weekly)	0.00	0.92	+28.4%
Wikipedia (monthly)	0.06	0.87	+16.9%
Weather (hourly)	0.77	0.69	+12.5%
Air quality (hourly)	0.34	0.84	+12.6%
Electricity (hourly)	0.47	0.83	-10.6%
Exchange rates (daily)	0.00	0.93	n.s.

Table 3: Series properties measured on the input window only, against the observed advantage. Neither column orders the last one.

Exchange rates (29 series, daily). The Friedman test does not reject. Every method, seasonal naive included, sits in one undifferentiated tied set. We include this group precisely so the benchmark can produce that answer: a design that cannot return "no difference" on a domain where no difference exists is not measuring what it claims to measure. The small panel widens the intervals, and we would not read the null as proof of no effect — but the point estimate for every model is also essentially the naive one.

4.3 The intrinsic explanation, and why we abandoned it

Ordering the groups by the gap between the best pretrained model and the best classical one gives a gradient: largest on Wikipedia traffic, smaller on weather and air quality, negative on electricity, null on exchange rates. The natural hypothesis is that some property of the series explains it, and the natural candidates are periodicity and noise — pretraining should help most where a repeating structure exists but is hard to estimate from a few hundred observations alone.

We tested that directly. For each group we measured, on the input window only, the median seasonal strength (from an STL decomposition at the group's seasonal period) and the median spectral entropy of the differenced series, both as defined in the feature literature used to characterise the M4 panel (Table 3). Neither explains the pattern. Seasonal strength is if anything *negatively* associated with the advantage: Wikipedia series have almost no measurable seasonality at the tested period and show the largest gains, while weather is the most strongly seasonal group in the study and shows a middling one. Spectral entropy fails in the other direction: exchange rates are as high-entropy as Wikipedia traffic and show no advantage at all. With seven groups no correlation here is statistically meaningful in any case, and we report the measurement rather than a story fitted to it.

4.4 Corpus familiarity survives the hold-out

What does line up with the gradient is not a property of the series but of the models' training data. TimesFM's authors describe its pretraining corpus as dominated by Wikipedia pageviews — on the order of 10^{11} time points at hourly, daily, weekly and monthly granularity, running to late 2023. Our three Wikipedia groups are pageviews at daily, weekly and monthly granularity, beginning in 2026. The hold-out is clean in time and identical in kind.

A domain being both the largest corpus component and the largest measured gain is suggestive, but it is one coincidence and admits a benign reading: perhaps Wikipedia traffic is simply a domain where all pretrained models do well. We can separate the two, because the pretrained models forecast the *same series*. Any property of those series — difficulty, seasonality, noise — applies equally to all of them and cancels in a within-family comparison. What does not cancel is what each model read during pretraining.

Ranking the six pretrained models against each other on every series, the TimesFM family sits -0.53 ranks below the Chronos family on Wikipedia and only -0.09 below it on the other four domains (1,500 vs. 754 series; Mann–Whitney $p < 10^{-5}$, rank-biserial -0.13). The model with Wikipedia in its corpus is specifically better at Wikipedia, and only there. The effect is modest in size, and we present it as such — but its direction is not something the series can account for.

We therefore read the gradient as a corpus-overlap effect rather than a capability one, and note that this is a hypothesis our design supports rather than a demonstrated mechanism. The decisive experiment is a benchmark whose domains are stratified by membership in each model's disclosed corpus, which requires disclosure the field does not currently provide. That, rather than a further increase in the size of the pretraining set, seems to us the useful next contribution.

4.5 What this means for evaluation

A temporal hold-out is necessary and not sufficient. It rules out the model having seen these observations; it does not rule out the model having spent its training on this kind of series, and the second effect is large enough to reorder a leaderboard. Reporting a single average across domains hides it entirely, because the average is dominated by whichever domains happen to be in the benchmark, and those are the well-published ones — which are also the ones most likely to be in a pretraining corpus.

4.6 Probabilistic forecasts, where the ordering changes

Point accuracy is only half of what these models produce, and the half that is usually reported. Every probabilistic method here also emits nine quantiles, from which we compute weighted quantile loss and the empirical coverage of the nominal 80% interval (Table 4). Two things appear that the point metrics hide.

Model	Wikipedia (daily)	Wikipedia (weekly)	Wikipedia (monthly)	Weather (hourly)	Air quality (hourly)	Electricity (hourly)	Exchange rates (daily)
Weighted quantile loss (lower is better)							
AutoARIMA	0.547	0.514	0.518	0.140	0.306	0.313	0.003
AutoETS	†	†	4.274	0.190	0.479	0.377	0.004
Chronos2	0.217	0.292	0.334	**0.122**	**0.260**	0.300	0.003
ChronosBoltBase	0.228	0.300	0.335	0.125	0.268	0.358	0.004
ChronosBoltSmall	0.233	0.300	0.340	0.125	0.264	0.365	0.004
LSTM	0.281	0.313	0.412	0.151	0.298	0.407	0.005
Moirai2	0.222	0.298	0.328	0.124	0.266	0.310	0.004
NBEATS	0.268	0.334	0.399	0.131	0.268	0.303	0.004
Theta	0.701	0.627	0.530	0.160	0.354	0.298	0.004
TimesFM	0.209	0.294	**0.327**	0.127	0.263	0.318	0.003
TimesFM3	**0.207**	**0.283**	0.329	0.127	0.262	**0.294**	**0.003**
80% interval coverage (nominal 0.80)							
AutoARIMA	0.96	0.94	0.91	0.72	0.82	0.74	0.88
AutoETS	0.94	0.95	0.88	0.80	0.89	0.83	0.88
Chronos2	0.76	0.72	0.70	0.73	0.74	0.65	0.75
ChronosBoltBase	0.79	0.75	0.73	0.73	0.72	0.62	0.76
ChronosBoltSmall	0.78	0.77	0.75	0.74	0.74	0.64	0.72
LSTM	0.84	0.75	0.86	0.76	0.80	0.61	0.63
Moirai2	0.74	0.71	0.70	0.74	0.73	0.68	0.66
NBEATS	0.78	0.69	0.78	0.78	0.82	0.65	0.55
Theta	0.97	0.94	0.93	0.85	0.83	0.86	0.87
TimesFM	0.82	0.77	0.70	0.76	0.78	0.65	0.75
TimesFM3	0.79	0.74	0.72	0.74	0.75	0.69	0.74

Table 4: Probabilistic accuracy and calibration. † marks a divergence (see §5), not a large loss. Coverage should be read against a nominal 0.80: below is overconfident, above is conservative.

First, the ranking by quantile loss largely agrees with the ranking by MASE, which is reassuring and not very interesting. Second, and less comfortably, **the calibration splits cleanly along family lines**. Every pretrained model's 80% interval covers less than 80% of outcomes — the worst, Moirai2, by 9 percentage points — while all three automatic classical methods cover *more* than nominal, Theta by 9 points. The pretrained models are systematically overconfident and the classical ones systematically cautious, consistently, across every domain in the study.

This matters more than the accuracy gap for anyone using a forecast to size a buffer, a reserve or an inventory. A model that is 10% better on MASE and whose 80% interval is really a 71% interval is not obviously the better tool for that job, and nothing in the point-accuracy tables that dominate this literature would tell you so. We report it without a proposed remedy: recalibration is well studied, and the finding here is simply that it is needed and is not being done.

4.7 Cost

The accuracy differences among the leading pretrained models are small enough to sit inside their rank intervals in most groups. Their costs are not comparable in the same way. An automatic ARIMA search costs a median 54× more per series than a pretrained forward pass, and that factor is not a constant: it runs from 5× on weekly series to 2,689× on weather (hourly), since the order search scales with series length and seasonal period while a forward pass does not. On the 7 groups where it completed, it was beaten on accuracy by the best pretrained model in 7. For a practitioner the

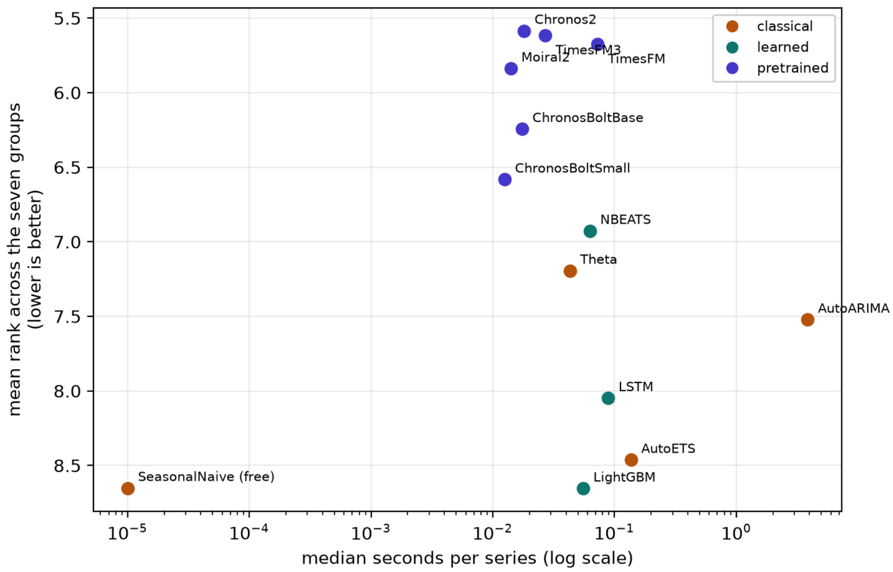

Figure 2: Cost against accuracy, aggregated over the seven groups. The pretrained models occupy a tight cluster at the top left — cheapest and most accurate — while the automatic ARIMA search sits two orders of magnitude to the right of them and ranks worse than a Theta baseline that costs a hundredth as much.

Model	Median s/series	Relative to ChronosBoltSmall
SeasonalNaive	0.0000	free
ChronosBoltSmall	0.0125	1.0×
Moirai2	0.0141	1.1×
ChronosBoltBase	0.0174	1.4×
Chronos2	0.0181	1.4×
TimesFM3	0.0269	2.2×
Theta	0.0429	3.4×
LightGBM	0.0550	4.4×
NBEATS	0.0627	5.0×
TimesFM	0.0726	5.8×
LSTM	0.0887	7.1×
AutoETS	0.1362	11×
AutoARIMA	3.8484	308×

Table 5: Median wall-clock seconds per series on identical hardware, including model load.

comparison that matters is therefore not among the pretrained models, whose differences are mostly not resolvable, but between that family and the automated classical search it would replace — and there the case is strongest exactly where the search is most expensive.

5 Failure modes we hit

Benchmark papers report the runs that worked. The ones that did not are more useful to the next person, and three of ours changed a number that would otherwise have been published.

A metric with no denominator. The Danish grid publishes a hydro-generation column for a price area with essentially no hydro. It is zero throughout, apart from a single 10^{-6} reading. MASE divides by the in-sample seasonal-naive error, which for that series is 10^{-9}, so the two neural models — which predicted a small positive quantity rather than exactly zero — scored 4×10^8 and 6×10^7. One series in forty-seven moved the group's mean MASE by six orders of magnitude. The rank-based conclusion for that group was unchanged, which is the argument for ranks made concrete rather than in the abstract. The series is now excluded by the rule in §3.2, and we flag that a relative threshold does not catch it: its scale is negligible in absolute terms and enormous relative to itself.

A classical baseline that diverges rather than degrades. AutoETS produces weighted quantile losses of order 10^{14} on two of the three Wikipedia groups — marked † in Table 4. This is not an artefact of our harness: the fitted model genuinely explodes on Wikipedia's spiky, heavy-tailed traffic, whose training ranges span three orders of magnitude within a single series, and it emits negative quantiles for a strictly non-negative quantity. We report it rather than silently substituting a damped variant, because an automatic method that fails this way on real data is a finding about the method. Its point forecasts on the same groups are poor but finite, which is why the failure is invisible in a MASE-only table.

Numerical precision on accelerator hardware. TimesFM raises a type error under float64 on Apple's Metal backend and must fall back to CPU for the affected series; Moirai-2 required the same treatment. Neither changes the forecasts, but both change the timing, and a cost table that did not disclose it would be comparing a GPU path against a CPU one without saying so. The timings in Table 5 are as measured, with the fallbacks in place, and `results/holdout_2026/timing.jsonl` records the device actually used for every cell.

What we checked rather than assumed. Only three models are re-run under multiple seeds, on the claim that the rest are deterministic. Since that claim would understate every significance test if it were wrong, `scripts/check_determinism.py` verifies it by running each supposedly deterministic model twice under different seeds and comparing forecasts exactly. Across the three stochastic models, the spread in mean MASE over three seeds is below 8% on every group.

6 Limitations

The corpus-overlap finding is an association, not a demonstrated mechanism. It rests on one corpus whose composition is publicly described and one domain that matches it; a model could be better at Wikipedia traffic for reasons unrelated to having read Wikipedia traffic, and our within-family comparison narrows that possibility without closing it. The effect is also modest — a rank-biserial correlation of -0.13 — and we resist reading a large practical consequence into it. Settling the question needs a benchmark stratified by corpus membership across several models, which in turn needs disclosure that most published corpora do not provide.

Beyond that: the hold-out is one forecast origin per group, where a rolling-origin evaluation would estimate variance across time as well as across series. Two domains contribute fewer than fifty series, so their rank intervals are wide, and the electricity result should be read as "no evidence of an advantage" rather than a demonstrated deficit. Pretraining cutoffs are taken from model documentation and cannot be independently verified; where a cutoff is undisclosed we substitute the release date. And the study covers five domains, more than the standard two but far from the diversity of real forecasting practice — with the caveat, central to our argument, that adding whichever domains are easiest to obtain is likely to add exactly the well-published ones that pretraining corpora already contain.

7 Conclusion

Evaluated on data that postdates them, time-series foundation models are strong on some domains, unnecessary on others, and useless where nothing works. We set out to explain that spread with a property of the series and could not: neither periodicity nor entropy orders it. What orders it is which domains the models were raised on. The largest advantage in our study falls on the domain that constitutes the bulk of one model's pretraining corpus, at the same granularities, and it is that model's family which holds the advantage there and nowhere else.

The practical consequence for benchmark design is that moving the test window forward is not enough. It is necessary — without it, nothing can be concluded at all — but it controls only for having seen these observations, not for having spent a hundred billion time points learning what this kind of observation looks like. A benchmark that wants to measure generalisation has to hold out domains, not just dates, and it can only do so if pretraining corpora are disclosed well enough to say which domains those are. The practitioner's version of the same point is shorter: before asking which model is best, ask whether your data looks like the data it was raised on.

We release the fetchers, the harness and the per-series losses so that both the negative result and the corpus effect can be tested on domains we did not consider.

8 Dataset documentation

What we release, and what we do not. We release the fetchers, not the data. Every group is rebuilt by a script in `data/sources/` that retrieves the observations directly from the publisher, with no credentials, and writes a gzipped CSV alongside a manifest recording the request parameters, the retrieval timestamp, the series count and a SHA-256 of the output. Redistributing the observations ourselves would add a copy that can drift from the source and would put us in the position of relicensing data we do not own; a fetcher plus a checksum gives a reader the same panel without either problem. The consequence is that a rebuild months from now returns a longer series than ours, which is why the forecast origin is fixed in the harness rather than implied by the file.

Sources and terms. Wikimedia pageviews come from the Wikimedia REST API; weather and air quality from Open-Meteo's ERA5 and CAMS archives; electricity from Energinet's Energi Data Service; exchange rates from the European Central Bank's SDMX API. All four are institutional open-data services that require no registration. Each publisher's own terms govern reuse of its data, and users of this benchmark are subject to them; the code we release is ours and carries the repository's licence.

Maintenance and drift. The panel is defined by a rule — the full city list, the full currency list, the publisher's own column set — and not by a frozen file, so it is stable under re-fetching except where a publisher adds or retires a series. The manifests make such a change visible rather than silent: a differing series count or checksum against the values we report is the signal that the panel has moved. We regard this as the correct trade for a benchmark whose entire premise is that the test window must keep moving forward. A frozen file would become contaminated the moment the next generation of models is trained on it, which is precisely the failure the paper documents.

Machine-readable metadata. The repository carries a Croissant record (`croissant.json`) describing all five sources, with the series count, date range and SHA-256 of each, generated from the fetchers' own manifests by `scripts/make_croissant.py` rather than written by hand — a hand-copied checksum is one that will be wrong after the next re-fetch.

Known limitations of the data. Two groups are small (46 and 29 series), which widens their rank intervals materially. The weather and air-quality panels are reanalysis products rather than station readings, so they are smoother than instrument data would be. The electricity panel covers one country's grid. And every domain here is one that somebody chose to publish continuously and openly, which is a biased sample of forecasting problems — biased, specifically, towards the kind of data that ends up in a pretraining corpus, which is a caveat that cuts against our own headline effect as much as for it.

Reproducibility

Every dataset is rebuilt by a script in `data/sources/` requiring no credentials. `scripts/run_holdout.sh` reproduces the runs, `scripts/significance.py` the tests, and `scripts/make_tables.py` every table and every number quoted above. Per-series losses are released alongside the aggregates so that any alternative test can be applied without re-running the models.

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